

Comparative water use by the riparian trees *Melaleuca argentea* and *Corymbia bella* in the wet–dry tropics of northern Australia

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Summary We examined sources of water and daily and seasonal water use patterns in two riparian tree species occupying contrasting niches within riparian zones throughout the wet–dry tropics of northern Australia: *Corymbia bella* Hill and Johnson is found along the top of the levee banks and *Melaleuca argentea* W. Fitzg. is restricted to riversides. Patterns of tree water use (sap flow) and leaf water potential were examined in four trees of each species at three locations along the Daly River in the Northern Territory. Predawn leaf water potential was higher than –0.5 MPa throughout the dry season in both species, but was lower at the end of the dry season than at the beginning of the dry season. Contrary to expectations, predawn leaf water potential was lower in *M. argentea* trees along the river than in *C. bella* trees along the levees. In contrast, midday leaf water potential was lower in the *C. bella* trees than in *M. argentea* trees. There were no seasonal differences in tree water use in either species. Daily water use was lower in *M. argentea* trees than in *C. bella* trees. Whole-tree hydraulic conductance, estimated from the slope of the relationship between leaf water potential and sap flow, did not differ between species. Xylem deuterium concentrations indicated that *M. argentea* trees along the riverbank were principally reliant on river water or shallow groundwater, whereas *C. bella* trees along the levee were reliant solely on soil water reserves. This study demonstrated strong gradients of tree water use within tropical riparian communities, with implications for estimating riparian water use requirements and for the management of groundwater resources.

Keywords: groundwater, hydraulic conductance, leaf water potential, sap flow.

Introduction

Vegetation along riverbanks protects water quality by filtering surface and subsurface water moving across and through the soil to the river and also helps to maintain stream water quality by regulating stream temperature (through shading), turbidity (through erosion control and debris trapping) and bank stabil-

ity (Askey-Dorin et al. 1999). Riparian communities are important sources of in-stream and terrestrial habitats (Catterall 1993), act as corridors for the movement of fauna through the landscape and often have higher biodiversity than surrounding ecosystems.

Riparian vegetation is sensitive to changes in hydrological regimes (Horton et al. 2001). Smith et al. (1991) found that riparian trees along reaches of streams in the Sierra Nevada subject to stream diversions had reduced leaf area, leaf thickness and significantly lower stomatal conductance and midday leaf water potential than similar species along undiverted reaches. Horton et al. (2001) reported that both *Populus fremontii* S. Wats. and *Salix gooddingii* Ball, two riparian trees in Arizona, had lower shoot water potentials, leaf gas exchange rates and increased canopy dieback during a dry year compared with a wet year. Riparian cottonwoods, in general, are very sensitive to altered water regimes. Rood et al. (2003) reviewed the ecophysiology of cottonwoods and concluded that, when alluvial groundwater is depleted, cottonwoods exhibit reduced predawn and midday leaf water potentials, stomatal conductance, transpiration and photosynthesis. These physiological responses are often accompanied by morphological changes including reduced shoot growth, altered root morphology, canopy dieback and, in extreme cases, increased mortality.

Dawson and Ehleringer (1991) demonstrated that riparian trees are not necessarily reliant on stream water. They found that riparian vegetation had access to a range of water sources within the riparian zone including recent rainfall, soil water, stream water and groundwater. Sources of water utilized were dependent on position in the landscape and age class (Dawson and Ehleringer 1991). Kolb et al. (1997) demonstrated that box elder (*Acer negundo* L.) trees rarely use stream water and derive most of their water requirements from deep soil water and recent precipitation, with recent precipitation being particularly important for young trees. In southern Australia, water sourced by *Eucalyptus camaldulensis* Dehn. on the Chowilla floodplain was dependent on position in the landscape and season. During the summer dry season, groundwater was a large component of transpiration (Thorburn et al. 1993). *Eucalyptus*

camaldulensis trees adjacent to the river used some river water, whereas trees further than 15 m from the river used no groundwater. During winter, *E. camaldulensis* used a combination of groundwater and recent precipitation (Mensforth et al. 1994).

Determining the water use requirements of riparian vegetation is complicated because of high spatial and temporal heterogeneity within the riparian zone. In Australia, little is known about the patterns of tree water use in tropical riparian systems. An understanding of this topic is important, however, because there is increasing pressure on natural resources throughout northern Australia, particularly surface water and groundwater, resulting from expansion of horticultural, pastoral and tourism industries. Major landscape-scale changes are occurring to ecosystems within the region and there is an immediate need to gain a better understanding of the water regimes of tropical ecosystems, so that water and landscape resource managers can manage existing water resources sustainably.

The principal aim of this study was to examine spatial and temporal patterns of water use in two riparian tree species, *Melaleuca argentea* W. Fitzg. and *Corymbia bella* Hill and Johnson, along the Daly River in northern Australia. The Daly River and its surrounding catchment are relatively undisturbed, providing a unique opportunity to study water use requirements of riparian vegetation before large-scale land use changes have impacted the hydrological cycle. Our specific objectives were to: (1) determine if water use in *C. bella* and *M. argentea* varies spatially and temporally; (2) determine if leaf water potential varies between the species, as well as between sites and during the dry season; (3) characterize the relationship between tree water use and leaf water potential in these species to determine if hydraulic conductance varies between species and sites; and (4) determine if *C. bella* and *M. argentea* access different water sources and if the source changes during the dry season.

Materials and methods

Site description

The Daly River is one of the Northern Territory's largest perennial rivers, with a catchment area of about 52,577 km² (Figure 1). The river is a significant biological, cultural and economic resource. Climate within the Daly River region is dominated by a distinct monsoonal wet season occurring from October to April followed by a prolonged dry season with little or no rainfall. Annual rainfall varies spatially within the catchment and ranges between 1000 and 1300 mm. Pan evaporation rates are high throughout the year, as are temperatures. Climatic variables for Wooliana, about 55 km downstream of the Douglas and Daly river confluence, are shown in Figure 2.

Catchment discharge is highly variable, reflecting the variable annual rainfall patterns within the region. Furthermore, there is strong connectivity between surface water and groundwater. As a result, groundwater levels vary annually and are principally a function of the amount of rainfall in the preced-

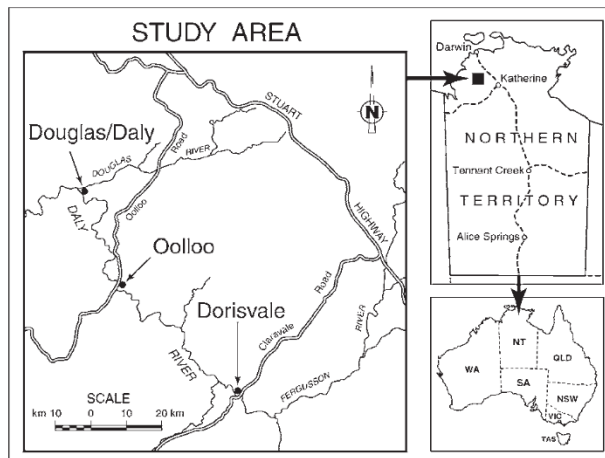


Figure 1. Location of the Daly River in northern Australia including sites where tree water use and soil and leaf water potentials were measured. Reproduced from Lamontagne et al. 2005.

ing wet seasons (P. Jolly, Northern Territory Department of Lands, Planning and Environment, unpublished data). During the wet season, a large proportion of water entering the river is derived from rainfall and surface runoff dominates river flow. However, during the dry season, base flow within the river is dominated by groundwater discharge from underlying limestone aquifers (Faulkes 1998).

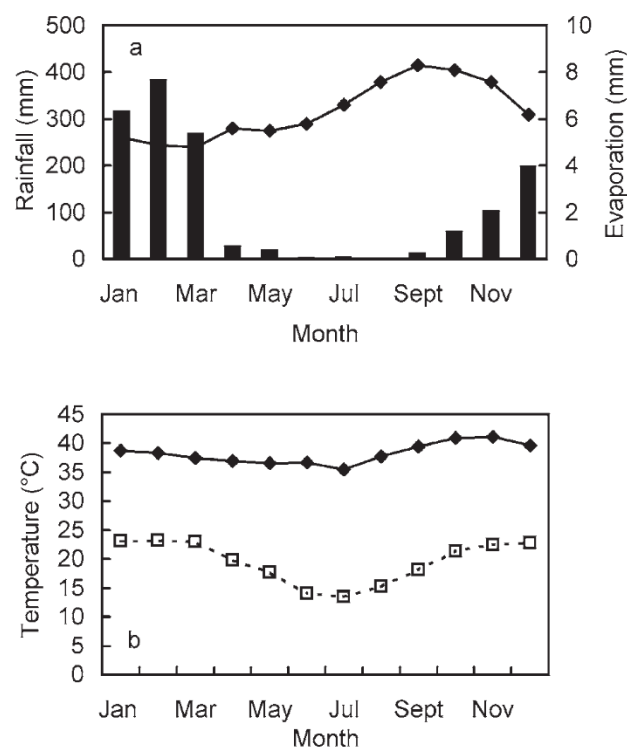


Figure 2. Patterns of (a) rainfall (bars) and pan evaporation (◆), (b) maximum temperatures (◆) and minimum temperatures (□) for Wooliana on the Daly River. Source: www.bom.gov.au (2005).

Riparian vegetation along the Daly River varies in width from a single line of trees to dense, closed forests along the levee banks and it contains many species typical of closed-monsoon forests (Faulkes 1998). Riverbanks are steep and rise in a series of terraces from the river. *Melaleuca argentea* and *Melaleuca leucadendra* L. trees occur on the lower river banks. Behind the strip of *Melaleuca* trees, closed-monsoon forest communities dominate river terraces. Open-forest communities occur along the levee banks (about 20 m above river base flow) and tend to be dominated by *Corymbia bella*; however, other species commonly found in savanna woodlands also occur. Faulkes (1998) has mapped in detail the cross-sectional vegetation profiles within all major rivers and tributaries in the Daly River catchment.

All plant water use and leaf water potential measurements were conducted at three sites along the Daly River: (1) Dorisvale crossing; (2) downstream from Oolloo crossing; and (3) at the confluence of the Douglas and Daly Rivers (Figure 1). These river sites were chosen to characterize spatial variability in tree water use along the river and because they could be accessed for most of the year, including the wet season when much of this remote river system is inaccessible.

Soil water availability

At each site, three replicate soil samples were collected with a hand auger (10 cm diameter) at 0.1, 0.5 and 1.0 m depths. Samples were collected at the end of the dry season (September 2000) and at the beginning of the next dry season (May 2001). Soil samples were stored in sealed containers and transported to the laboratory in insulated containers. Soil matric potential (Ψ_s) was measured by the filter paper technique (Greacen et al. 1989). Briefly, Whatman 42 filter papers were dried, weighed, placed in between two more filter papers and sandwiched within the soil sample. The water potential of the papers was allowed to equilibrate with Ψ_s over a 7-day period at a constant temperature of 25 °C. Soil matric potential was then calculated as the relationship between the relative water content (RWC) of the middle filter papers and the matric potential of the filter paper according to Equations 1 and 2:

$$\Psi_s = -\exp(12.27 - 17.93\text{RWC}) \quad (1)$$

for $0.278 < \text{RWC} < 0.453$

$$\Psi_s = -\exp(5.55 - 3.11\text{RWC}) \quad (2)$$

for $0.453 < \text{RWC} < 1.784$

A weighted mean Ψ_s (Hacke and Sperry 2002) was calculated for each site to facilitate comparisons with leaf water potential.

Leaf water potential

On each sampling date, leaf water potential was measured on three leaves from each of four trees per species at all sites. Measurements of leaf water potential were conducted in July–August 2000 (dry season), September–October 2000 (end of dry season) and in May 2001 (beginning of dry season). Measurements commenced predawn and were thereafter conduct-

ed at about hourly intervals until midday. Leaves were harvested from trees for which tree water use was being concurrently estimated. Harvested leaves were immediately placed in humid bags that were sealed and stored in a dark, insulated container to ensure that leaves were not rapidly transpiring during measurement (Turner and Long 1980). All measurements were made within 30 min of collection with a Scholander type pressure chamber (Soil Moisture Equipment Corporation, Santa Barbara, CA). Although it would have been preferable to measure water potentials on covered or shaded leaves (Meinzer et al. 2001, David et al. 2004), the canopies of these species are quite sparse and so selecting shaded leaves that were not actively transpiring was not feasible. In addition, canopy heights and steep riverbanks severely restricted access to the canopy.

Isotopic analysis

Deuterium concentrations were measured in xylem water and river water samples at Dorisvale and Oolloo in May and September 2000. River water was sampled, because groundwater is a major component of river base flow in the Daly River during the dry season. Deuterium signatures of river water and groundwater are indistinguishable along the Daly River during the dry season (Lamontagne et al. 2005). Water was collected from near the riverbank at a depth of about 0.5 m.

At each site, three twigs were collected from three trees per species for analysis of deuterium concentrations in xylem sap. Twigs were about 10–25 mm in diameter and were cut with pruning shears and immediately placed in glass airtight sample bottles and covered with kerosene. Isotope analyses were conducted by CSIRO Land and Water in Adelaide, South Australia, as described in detail by Lamontagne et al. (2005).

Tree water use

Tree water use was examined in each of four *C. bella* and *M. argentea* trees at each site (Figure 1). Sampling was conducted during the dry season (July–August 2000), at the end of the dry season (September–October 2000) and at the beginning of the next dry season (May 2001). Tree water use was estimated with commercially available sap flow loggers (Green-span Sap flow, Greenspan Technology Warwick QLD, Australia). Sap velocity was measured radially across the sapwood at four points. A 1.8-s heat pulse was fired at 15-min intervals over a 5-day period in each tree at each site during each of the sampling periods. Heat pulse velocity was converted to sap velocity following Edwards and Warwick (1984). Swanson and Whitfield (1981) solutions were used to correct heat pulse velocity for the effects of wounding (wound diameter = 3.1 mm). Sapwood area was determined by taking cores from each tree, and sapwood was distinguished from heartwood on the basis of a distinct colour change. These cores were used for determination of the volumetric wood and water contents of each sampled tree. Sap velocity was scaled to tree water use by the weighted averages technique (Hatton et al. 1990). For each tree, diameter at 1.3 m was recorded and bark thickness and depth of sapwood were also recorded.

Comparisons between water use expressed on a sapwood

area basis and leaf water potential were made by taking the mean of 15-min fluxes from before and after the time leaves were sampled for leaf water potential. The slope of the relationship between water use and leaf water potential was used as an estimate of hydraulic conductance, expressed in units of $\text{kg s}^{-1} \text{m}^{-2} \text{MPa}^{-1}$ to facilitate comparison with previously reported values of hydraulic conductance. We also used the relationship between sap flow and leaf water potential to predict the leaf water potential at zero flow (Bucci et al. 2004, O'Grady et al. 2006).

Results

Soil water availability

Soil matric potentials for each site are shown in Table 1. Soil matric potential in the surface soils was highly variable and generally low. Despite this variability, several trends were evident. Soil matric potential increased with soil depth. Matric potential was lower in levee soils than in riverbank soils in both seasons and was higher at the beginning of the dry season (May 2001) than at the end of the dry season (October 2000).

Leaf water potential

Mean Ψ_s , predawn leaf water potential and predicted leaf water potential at zero sap flow for each species are presented in Table 2. Results of the analysis of variance for predawn and midday leaf water potentials are shown in Table 3. Predawn leaf water potential was higher than -0.5 MPa in both species at all sites at all times with the exception of *M. argentea* trees at the Dorisvale site in July 2000. These trees had a lower predawn leaf water potential (-0.68 MPa) than *M. argentea* trees at either Oolloo or the Douglas/Daly confluence (-0.48 and -0.33 MPa, respectively). Predawn leaf water potential was slightly but significantly higher in *C. bella* trees along the lev-

ees than in *M. argentea* trees along the river (-0.29 versus -0.38 MPa, ANOVA $F = 26.99$, $df = 1.18$, $P < 0.01$). Furthermore, predawn leaf water potential in *C. bella* was lower at the end of the dry season (-0.35 MPa) than during the dry season (-0.24 MPa) or at the end of the wet season (-0.25 MPa). In contrast, predawn leaf water potential in *M. argentea* was lower during the dry season (-0.49 MPa) than at the end of the dry season (-0.32 MPa) or at the end of the wet season (-0.35 MPa).

For all trees, there was predawn disequilibrium between Ψ_s and leaf water potential. Predawn leaf water potentials predicted from extrapolations of the linear relationship between sap flow and leaf water potential were higher than measured predawn leaf water potentials (Table 2). Both measured and predicted predawn leaf water potentials were higher than the weighted mean Ψ_s in the top meter of soil. The predicted values of predawn leaf water potential suggest that roots must have been taking up water from the soil profile at soil matric potentials close to zero.

Midday leaf water potential was significantly lower than predawn leaf water potential (Figure 3). Generally, midday leaf water potential was lower in *C. bella* trees along the levees than in *M. argentea* trees along the riverside (-1.7 versus -1.4 MPa). The exception was *M. argentea* trees at Dorisvale during May 2001 (-1.7 and -1.2 MPa for *M. argentea* and *C. bella*, respectively). For both species, midday leaf water potential was lower at the end of the dry season (October 2000, -1.8 and -1.5 MPa for *C. bella* and *M. argentea*, respectively) than at the end of the wet season (May 2001, -1.6 and -1.4 MPa for *C. bella* and *M. argentea*, respectively) or during the dry season (July 2000, -1.7 and -1.2 MPa for *C. bella* and *M. argentea*, respectively). Consequently, the difference between predawn and midday leaf water potential was larger in *C. bella* trees than in *M. argentea* trees and was larger during

Table 1. Mean ($\pm$ SE) soil matric potentials (MPa) at three sites along the Daly River at the end of the dry season and at the end of the wet season.

Date	Site	Depth (m)	Levee	River
September 2000	Dorisvale	0.1	-41.6 ± 1.6	-89.2 ± 7.6
		0.5	-10.4 ± 4.5	-7.4 ± 6.3
		1.0	-3.3 ± 2.6	-1.2 ± 1.2
	Oolloo	0.1	-125.7 ± 43.9	0.0 ± 0.0
		0.5	-24.8 ± 9.2	0.0 ± 0.0
		1.0	-4.0 ± 2.4	-0.6 ± 0.6
	Douglas/Daly	0.1	-48.8 ± 1.6	-42.2 ± 0.0
		0.5	-39.8 ± 4.6	-2.5 ± 0.3
		1.0	-27.6 ± 13.6	-3.3 ± 0.6
May 2001	Dorisvale	0.1	-55.2 ± 21.5	-1.0 ± 0.8
		0.5	-3.3 ± 2.4	-5.5 ± 2.7
		1.0	-0.5 ± 0.4	-0.4 ± 1.8
	Oolloo	0.1	-65.9 ± 5.4	-19.8 ± 0.0
		0.5	-10.4 ± 2.8	-8.5 ± 0.0
		1.0	-4.3 ± 1.2	-1.2 ± 0.5
	Douglas/Daly	0.1	-16.7 ± 8.0	-1.9 ± 0.9
		0.5	-9.5 ± 3.1	-2.3 ± 1.0
		1.0	-6.4 ± 1.1	-6.9 ± 5.1

Table 2. Comparisons of soil matric potentials (Ψ_s) with measured predawn leaf water potentials ($\Psi_{pd \text{ (meas)}}$) and predicted zero flow leaf water potentials ($\Psi_{pd \text{ (pred)}}$). Soil matric potential is the weighted mean soil matric potential of the top meter of soil. Measured predawn leaf water potential is predawn leaf water potential measured with a pressure chamber and $\Psi_{pd \text{ (pred)}}$ is the water potential estimated from the sap flow and leaf water potential relationship. Abbreviations: DV = Dorisvale crossing; OL = Oolloo crossing; and DD = Douglas/Daly.

Date	Site	<i>Corymbia bella</i>			<i>Melaleuca argentea</i>		
		Ψ_s (MPa)	$\Psi_{pd \text{ (meas)}}$ (MPa)	$\Psi_{pd \text{ (pred)}}$ (MPa)	Ψ_s (MPa)	$\Psi_{pd \text{ (meas)}}$ (MPa)	$\Psi_{pd \text{ (pred)}}$ (MPa)
Sept 2000	DV	-12.7	-0.32	-0.18	-18.2	-0.25	-0.08
	OL	-33.3	-0.34	-0.26	-0.2	-0.29	-0.09
	DD	-35.7	-0.38	-0.21	-8.9	-0.41	-0.27
May 2001	DV	-9.3	-0.25	-0.20	-3.6	-0.34	-0.16
	OL	-16.9	-0.34	-0.06	-7.6	-0.42	-0.26
	DD	-10.5	-0.16	-0.13	-2.9	-0.28	-0.12

the dry season than during the wet season (Table 4) for both species.

Isotopic analysis

River water and xylem deuterium concentrations are shown in Table 5. Deuterium concentrations in the *M. argentea* trees were similar to the concentration of deuterium in river water at both sites and sampling times. However, deuterium concentrations of *M. argentea* trees and river water were more enriched than the xylem deuterium concentration of *C. bella* trees. Furthermore, at both sites, deuterium concentrations in *C. bella* trees were more enriched at the end of the dry season (September 2000) than at the end of the wet season (May 2000), whereas deuterium concentrations in *M. argentea* did not change seasonally.

Tree water use

Tree water use along the Daly River varied with species, time of day and tree size. In both species, tree water use was positively correlated with tree size; however, the relationship was more variable in *M. argentea* trees than in *C. bella* trees (Figure 4). Daily tree water use was lower in the *M. argentea* trees along the river than in the *C. bella* trees along the levee (ANCOVA $F = 9.08$, $df = 1.31$, $P < 0.01$). There were no significant differences between sites or seasons. When normalized by sapwood area, water use was lower in *M. argentea*

trees than in *C. bella* trees. The exceptions to this were in August 2000 (2833 and 2581 $\text{kg m}^{-2} \text{day}^{-1}$ for *M. argentea* and *C. bella*, respectively) at the Douglas/Daly site and in May 2001 at the Oolloo site (4282 and 3629 $\text{kg m}^{-2} \text{day}^{-1}$ for *M. argentea* and *C. bella*, respectively). Mean daily water use in *C. bella* and *M. argentea* was 4278 and 3244 $\text{kg m}^{-2} \text{day}^{-1}$, respectively. A detailed summary of seasonal water use in *C. bella* and *M. argentea* trees is shown in Table 6.

Water use and leaf water potential

There were strong linear relationships between tree water use and leaf water potential in both species (Figure 5). There was no significant difference in hydraulic conductance between dates, sites or species. Hydraulic conductance was, however, variable. A detailed summary of hydraulic conductance in *C. bella* and *M. argentea* trees is shown in Table 7. Mean whole-tree hydraulic conductance was 0.056 and 0.069 $\text{kg s}^{-1} \text{m}^{-2} \text{MPa}^{-1}$ in *M. argentea* and *C. bella*, respectively.

Discussion

Leaf water potential

Predawn leaf water potential was generally higher than -0.5 MPa year-round, indicating that neither of the riparian tree species developed significant water stress during the dry season, despite low soil matric potentials in the top meter of

Table 3. Statistical model and ANOVA results for predawn (Ψ_{pd}) and midday (Ψ_{md}) leaf water potentials at sites along the Daly River. Symbols: *** = $P < 0.01$; and ** = $P < 0.05$. Abbreviation: df = degrees of freedom.

Factor	df	Ψ_{pd}	Ψ_{md}
Date	2	2.41 ***	8.01 ***
Site	2	4.54 ***	2.13
Species	1	26.99 ***	51.80 ***
Date $\times$ site	4	7.12 ***	1.74
Date $\times$ species	2	6.38 ***	2.71
Site $\times$ species	2	1.72 **	25.68 ***
Date $\times$ site $\times$ species	4	1.06	6.41 ***
Tree	18	1.99 **	1.24
Residual	129	—	—

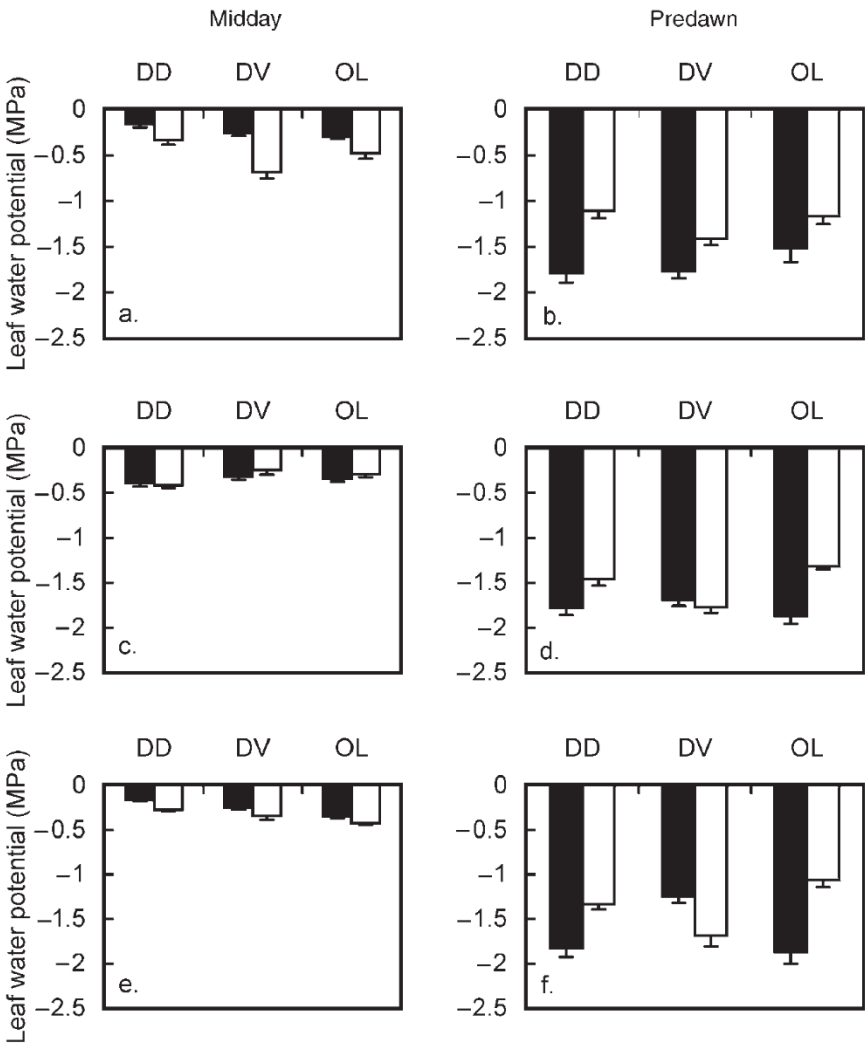


Figure 3. Leaf water potentials of *C. bella* and *M. argentea* trees in July–August 2000 (a, b), October 2000 (c, d) and May 2001 (e, f) at three sites along the Daly River. Filled bars represent the mean $\pm$ SE for *C. bella* and open bars represent the mean $\pm$ SE for *M. argentea*. Abbreviations: DD = Douglas/Daly; DV = Dorisvale; and OL = Oolloo.

soil. High year-round predawn leaf water potential is indicative of access to reliable sources of water such as groundwater or river water. Drake and Franks (2003) found that predawn leaf water potential in five riparian tree species in northern Queensland remained high throughout the year, but noted that there was a significant drop at the end of the dry season in two

species that lacked access to groundwater or river water. We found that predawn leaf water potentials were lower in both species at the end of the dry season. However, when leaf water potentials calculated for zero sap flow were considered, values closer to zero were obtained at the beginning of the dry season and at the end of the dry season for both species (Table 2). Don-

Table 4. Differences between predawn and midday leaf water potentials (MPa) for *C. bella* and *M. argentea* trees at three sites along the Daly River. Abbreviations: DD = Douglas/Daly; DV = Dorisvale crossing; and Oolloo = Oolloo crossing. Values are means $\pm$ SE.

Date	Site	<i>C. bella</i>	<i>M. argentea</i>
August 2000	DD	-1.45 ± 0.25	-0.72 ± 0.30
	DV	-1.51 ± 0.07	-0.71 ± 0.09
	Oolloo	-1.25 ± 0.15	-0.67 ± 0.05
October 2000	DD	-1.37 ± 0.07	-0.66 ± 0.19
	DV	-1.42 ± 0.04	-1.54 ± 0.07
	Oolloo	-1.47 ± 0.12	-1.09 ± 0.06
May 2001	DD	-1.21 ± 0.23	-0.63 ± 0.21
	DV	-1.00 ± 0.07	-1.27 ± 0.17
	Oolloo	-0.95 ± 0.25	-0.36 ± 0.18

Table 5. Deuterium concentration (‰) in *M. argentea*, *C. bella* and the Daly River at Oolloo and Dorisvale in May 2000 and September 2000. Data represent means $\pm$ SE of three samples unless otherwise noted (number of samples in parentheses).

Site	Date	<i>M. argentea</i> (‰)	<i>C. bella</i> (‰)	Daly River (‰)
Dorisvale	May 2000	-55.9 ± 1.7	-100.1 ± 2.4	-52.5 ± 0.1 (2)
	September 2000	-56.4 ± 1.5	-85.3 ± 2.9	-45.4
Oolloo	May 2000	-45.2 ± 0.8	-70.0 ± 5.0	-46.4 ± 1.1 (6)
	September 2000	-45.6 ± 1.4	-63.3 ± 3.8	-44.2 ± 0.4 (5)

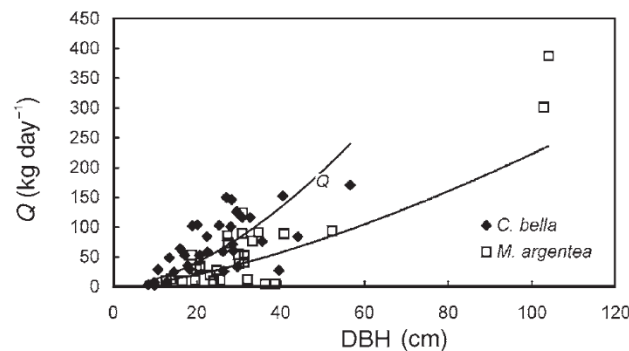


Figure 4. Relationship between diameter at breast height (DBH) and water use (Q) for *C. bella* and *M. argentea* trees along the Daly River. Data were pooled for sites and seasons and each value is the mean daily water use of each tree. Power relationships are shown to highlight differences in water use for a given DBH between species: *C. bella*: $y = 0.26x^{1.69}$, $r^2 = 0.56$; and *M. argentea*: $y = 0.25x^{1.47}$, $r^2 = 0.38$.

ovan et al. (2001) reported that predawn disequilibrium was common and proposed nocturnal transpiration as a mechanism contributing to this disequilibrium. Nighttime transpiration has been recorded in many species (see Benyon 1999, Snyder et al. 2003, Bucci et al. 2004). Bucci et al. (2004) found that predawn leaf water potential of exposed leaves was lower than that of covered leaves. They demonstrated that extrapolation of the relationship between sap flow and leaf water potential of uncovered leaves yielded leaf water potentials at zero sap flow similar to those of covered leaves.

Midday leaf water potentials of *C. bella* and *M. argentea* trees were higher than the midday leaf water potentials reported for tree species in the savannas near Darwin. Minimum leaf water potentials were less than -2 MPa for five evergreen species in the wet tropical savanna during the dry season (Myers et al. 1997). Midday leaf water potential was lower in *C. bella* than in *M. argentea*. Furthermore, for both species, midday leaf water potential was lower at the end of the dry season than at the beginning of the dry season. Differences in midday leaf water potential between sampling dates were small. For many species, the minimum leaf water potential is close to the threshold cavitation water potential (Cochard et al. 1996, Sperry 2000). Brodribb and Holbrook (2003) found that once this threshold was reached, significant loss of leaf conductance occurred over a narrow range in four tropical dry forest species. Loss of conductance correlated with reduced stomatal conductance and photosynthetic capacity (Brodribb and Holbrook 2003). If water deficits are persistent, they may lead to longer-term morphological adjustments such as increased root:shoot ratio (Li and Cohen 2005), reduced sapwood area to leaf area ratio (Sperry et al. 2002) or even increased mortality (Cooper et al. 2003).

Tree water use and hydraulic conductance

Tree water use and hydraulic conductance did not vary between sites. However, large variability and low replication (four trees per species per site) between sites may have masked differences. There were strong relationships between tree diameter and water use at each site (Figure 4). These types of relationships form the basis for scaling water use in forests. Hatton et al. (1998) and Hutley et al. (2000) used a universal

Table 6. Seasonal water use ($\text{kg day}^{-1} \text{m}^{-2}$) normalized by sapwood area for *C. bella* and *M. argentea* trees along the Daly River. Data represent means $\pm$ SE. Abbreviations: DD = Douglas/Daly; DV = Dorisvale crossing; and Oolloo = Oolloo crossing.

Season	Site	<i>C. bella</i>	<i>M. argentea</i>
August 2000	DD	2581.2 ± 503.2	2833.5 ± 1576.9
	DV	5280.0 ± 1548.9	5065.2 ± 1088.2
	Oolloo	5177.2 ± 1425.9	2776.8 ± 565.8
October 2000	DD	2864.0 ± 1211.7	1526.2 ± 937.8
	DV	6116.7 ± 642.0	4763.9 ± 1603.8
	Oolloo	5481.9 ± 1873.6	4944.9 ± 3308.3
May 2001	DD	4123.0 ± 1567.2	1819.3 ± 565.1
	DV	2935.1 ± 610.4	2242.1 ± 933.5
	Oolloo	3628.8 ± 590.9	4282.1 ± 1513.8

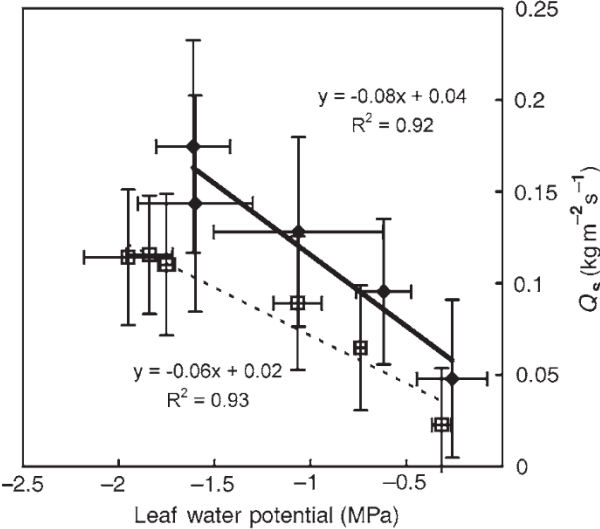


Figure 5. Example of the relationship between sap flow (Q_s) and leaf water potential in *C. bella* (◆) and *M. argentea* (□) at the Dorisvale crossing site at the end of the dry season, September 2000. Values are means $\pm$ SE of four trees.

relationship to scale canopy water use in the savannas near Darwin; however, a universal relationship is not likely to be useful in these riparian forests because water use was lower in *M. argentea* than in *C. bella*. Riparian forests are floristically and structurally complex, with over forty tree species recorded in the forests along the Daly River (O’Grady et al. 2002). Estimation of stand water use in these forests will require careful consideration of this structural and floristic complexity.

Tree water use at the beginning of the dry season was similar to water use at the end of the dry season in both species, even though Ψ_s declined over the dry season. O’Grady et al. (1999) and Eamus et al. (2000) found that some savanna species maintain a constant rate of water use throughout the dry season, despite declining predawn leaf water potential and increasing vapor pressure deficit. In two savanna eucalypts, leaf area declined during the dry season as predawn leaf water potential decreased, consistent with an adjustment in sapwood area to leaf area ratio to maintain a constant water potential gradient within the tree (Whitehead 1998). Because the decline in leaf

area was proportional to the increase in vapor pressure deficit, water use by the two savanna eucalypts remained constant over the dry season (O’Grady et al. 1999). It is unlikely that our riparian trees use the same mechanism to regulate transpiration. In our study, the difference between predawn water potential and midday water potential was only slightly larger at the end of the dry season than at the beginning of the dry season. Furthermore, water use was higher, although not significantly so, at the end of the dry season than at the beginning of the dry season. Under well-watered conditions, water use will continue at the rate imposed by vapor pressure deficit (Jarvis 1993), as long as the water potential gradient can be maintained within safe limits (Whitehead 1998, Sperry 2000).

Hydraulic conductance of both species in this study ($0.06 \text{ kg m}^{-2} \text{ s}^{-1} \text{ MPa}^{-1}$) was higher than dry season conductance of the savanna species, *E. tetradonta* F. Muell., near Darwin, $0.03 \text{ kg m}^{-2} \text{ s}^{-1} \text{ MPa}^{-1}$; Prior 1997) and higher than the dry-season conductance of woodland *Quercus rotundifolia* Lam. in Portugal ($0.04 \text{ kg s}^{-1} \text{ m}^{-2} \text{ MPa}^{-1}$; David et al. 2004). High conductance is a condition of high productivity (Tyree 2003) and helps to explain the higher basal area (BA) and leaf area index (LAI) of riparian forests (BA = $71 \text{ m}^2 \text{ ha}^{-1}$; O’Grady et al. 2002) compared with the surrounding savanna ecosystems (BA = $9 \text{ m}^2 \text{ ha}^{-1}$, LAI = 0.8; O’Grady et al. 2000).

Hydraulic conductance was highly variable, suggesting that calculating hydraulic conductance from the slope of the relationship between sap flow and water potential is subject to error. Both individual water use on a sapwood area basis and leaf water potential measurements were highly variable, even though care was taken to ensure that leaves were immediately bagged and placed in dark containers to ensure that the water potential did not change during the measurement (Turner and Long 1980). Measuring the water potential of covered non-transpiring leaves (xylem potential) would have reduced the variability in our water potential estimates (Turner and Long 1980, Meinzer et al. 2001); however, logistical constraints (canopy height, steep river banks) made this approach impractical. Bucci et al. (2004) demonstrated that the difference between leaf water potential of covered and uncovered leaves increased as transpiration increased throughout the morning in three species in Brazil, suggesting that hydraulic conductance calculated from the slope of the relationship between sap flow and leaf water potential of uncovered leaves is likely to be

Table 7. Hydraulic conductance ($\text{kg m}^{-2} \text{ s}^{-1} \text{ MPa}^{-1}$) calculated from the slope of the relationship between flux (normalized by sapwood area) and leaf water potential for *C. bella* and *M. argentea* along the Daly River. Data are means $\pm$ SE. Abbreviations: DD = Douglas/Daly; DV = Dorisvale crossing; and Oolloo = Oolloo crossing.

Date	Site	<i>C. bella</i>	<i>M. argentea</i>
October 2000	DD	0.02 ± 0.01	0.06 ± 0.02
	DV	0.08 ± 0.01	0.06 ± 0.01
	Oolloo	0.08 ± 0.03	0.12 ± 0.08
May 2001	DD	0.08 ± 0.03	0.04 ± 0.01
	DV	0.04 ± 0.02	0.03 ± 0.01
	Oolloo	0.04 ± 0.01	0.12 ± 0.04

overestimated slightly. Calculation of hydraulic conductance from the slope of the relationship between sap flow and leaf water potential assumes a steady-state condition (Li et al. 2005). However, capacitance may contribute to water use and result in nonlinear relationships between sap flow and leaf water potential. In our study, the relationships between sap flow and leaf water potential were linear even though sap flow was measured at the base of the tree, which would maximize capacitance-induced lags between transpiration and flow.

Sources of water

Comparison of predawn leaf water potential with the weighted mean Ψ_s in the top meter of soil demonstrated that both species were accessing water sources deeper in the soil profile. The low soil matric potentials in the top meter were unexpected, especially at the beginning of the dry season. Lamontagne et al. (2005) also reported low water potentials in surface soils along the Daly River (less than -4 MPa). High evaporation rates during both the wet and dry seasons combined with rapid drainage in these sandy soils probably contributed to the low values. Hutley et al. (2000) demonstrated that understory and soil evaporation were major components of the total evapotranspiration during the wet season in savannas and this would contribute to a rapid drying of the surface soils.

Along the Daly River, there is no isotopic discrimination between groundwater and river water (Lamontagne et al. 2005). Deuterium concentrations in xylem of *M. argentea* trees along the river were similar to the deuterium concentration in the river water (Table 4). Furthermore, Ψ_s in the top meter was generally lower than both predawn and midday leaf water potentials indicating that these trees were reliant solely on river water or groundwater. Graham (2001) demonstrated that sap flow in roots of *M. argentea* trees in the Pilbara (NW Western Australia) with direct access to river water was higher than in roots in dry river bank soils, indicating that *M. argentea* trees were sourcing most of their water directly from the river.

Deuterium signatures in xylem of *C. bella* trees were lower than those of river water or groundwater throughout the dry season, indicating that *C. bella* trees were reliant on soil water. Soil water potential in the top meter, however, was lower than both predawn and midday leaf water potentials, indicating that soil water was unavailable in the top meter during the dry season. These conflicting results suggest that *C. bella* trees are reliant on deep soil water reserves. Soil matric potential in May 2001 was about -0.5 MPa at 3 m at Dorisvale and Ooloo (data not shown), suggesting that water below this depth would have been available to trees for transpiration. Lamontagne et al. (2005) demonstrated that deuterium became more enriched as depth to groundwater increased. Xylem deuterium concentrations in *C. bella* were more enriched in September compared with May, suggesting increased reliance on deeper soil water or groundwater as the dry season progressed.

In conclusion, there was little seasonality in tree water use in the two tree species studied along the Daly River in the Northern Territory. Predawn leaf water potential remained high throughout the year, despite a prolonged dry season and low

soil matric potentials in the top meter, suggesting that both species had adequate access to deep soil water, groundwater or river water. Xylem deuterium concentrations in *M. argentea* were similar to river water. For *C. bella*, the xylem deuterium signature was consistent with access to deep soil water. Predawn leaf water potential and tree water use in *M. argentea* trees were lower than predawn leaf water potential and water use in *C. bella* trees, suggesting that there were strong gradients in water use within the riparian zone. High year-round water availability in these habitats and access to deep soil water and groundwater make these communities vulnerable to changes in hydrological regimes resulting from land clearing or groundwater extraction.

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